

Hiroshi Madrigal Hojo

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EDUCATION

University of the Philippines Diliman Bachelor of Science in Physics

Quezon City, Philippines
Expected July 2027

- University Scholar Award (Academic Year 2024–2025)
- Undergraduate Student, Condensed Matter Physics Laboratory — Semiconductor Group, National Institute of Physics

Ateneo de Manila University

Quezon City, Philippines

Elementary through Senior High School (STEM track), 2010–2022

RESEARCH EXPERIENCE

Undergraduate Researcher

January 2025 – Present

Condensed Matter Physics Laboratory — Semiconductor Group, National Institute of Physics, UP Diliman

- Two years of hands-on cleanroom and characterization experience: sample fabrication (electrochemical anodization, spray pyrolysis, hydrothermal synthesis) and structural/optical characterization (XRD, Raman, UV-Vis, and PL spectroscopy).
- Python-based data processing and analysis of spectroscopic measurements.

UNDERGRADUATE THESIS

Photoluminescence of Zinc Oxide on Porous Silicon Grown via Hydrothermal Synthesis

June 2026

Adviser: Dr. Armando S. Somintac — National Institute of Physics, UP Diliman

Fabricated PSi substrates and ZnO nanostructures; demonstrated that hydrothermally grown ZnO on PSi, upon annealing at 450 °C, produces broadband visible emission spanning 400–900 nm with resolved blue, green, and red components, establishing a silicon-compatible platform for white-light-emitting devices. Characterized using XRD, Raman spectroscopy, and PL spectroscopy.

CONFERENCE PROCEEDINGS

Hojo, H. M.*, **Husay, H. A. F.**, et al.. “Synthesis and characterization of broadband white light emission from ZnO nanostructures on porous silicon.” *Proceedings of the Samahang Pisika ng Pilipinas (SPP) 2026*, 2026. [\[link\]](#)

Manuel, B. A. J.*, **Hojo, H. M.**, et al.. “A comparative study of the structural and optical properties of ZnO films synthesized through spray pyrolysis and hydrothermal growth.” *Proceedings of the Samahang Pisika ng Pilipinas (SPP) 2026*, 2026. [\[link\]](#)

Tomambo, S. R. D.*, **Hojo, H. M.**, et al.. “Influence of silver immersion time on the SERS activity of silver nanostructures on porous silicon templates for methylene blue detection.” *Proceedings of the Samahang Pisika ng Pilipinas (SPP) 2026*, 2026. [\[link\]](#)

Husay, H.*, **Hojo, H.**, et al.. “Determining the resistivity of aluminum-doped zinc oxide films through current-voltage measurements.” *Proceedings of the Samahang Pisika ng Pilipinas (SPP) 2025*, 2025. [\[link\]](#)

Lopez, J.* , Hojo, H., et al.. “Comparative Raman analysis of porous silicon DBRs before and after lift-off under visible and near-infrared laser excitation.” *Proceedings of the Samahang Pisika ng Pilipinas (SPP) 2025*, 2025. [\[link\]](#)

*Corresponding author

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Co-Founder & Editor-in-Chief, [Riles Magazine](#)

January 2026 – Present

- Co-founded an independent local digital magazine from scratch.
- Built the website from scratch using HTML and CSS.
- Editorial direction, layout design (Adobe Illustrator, Adobe Photoshop, Canva), and Digital content strategy.

1st Runner-Up — Blue Case Competition (Mind the Gap) 2026

April 2026

Ateneo Blue Consulting Group • Prize: P20,000 • Toyota Valenzuela Inc.

- Strategized a recovery plan for aftersales revenue leakage across the local franchise network. By prioritizing customer reintegration and trust-building initiatives, the project aims to bridge the 'trust gap' and elevate client retention from 50% to 68% over five years.

UP Advertising Core

2024 – 2026

- **Creatives Manager:** Creative Direction, Design, Publications
2024: AdSchool 18.2, AdCore Week 18.2, JuiceLetter 19.1
2025: TeenSpeak 19.2
- **Strategic Execution and Logistics Manager:** Transportation, Procurement, Sourcing, Event Operations
- **2026:** AdCore Week 20.2

College of Science Student Council, UP Diliman

2023 – 2024

- **Councilor** (2023–2024)
Managed and produced visual communication for the council, online and on-campus
Co-organized internal College of Science events
Unified and provided aid to College of Science organizations
- **Volunteer** (2022–2023)
Elected Vice-chairperson of the (now dissolved) College of Science Freshies, Shiftees, and Transferees (FST) Council
Handled Internal Affairs of the Council
Spearheaded Mental Health Awareness Month initiatives

Project Lead — Iskrash Course (Freshmen Vlog Series), UP Diliman

2022

- Managed a vlog series showcasing UP Diliman culture for incoming freshmen; see facebook.com/IskrashCourse.

SKILLS & TECHNICAL COMPETENCIES

Programming	Python (data processing, analysis, visualization) • L ^A T _E X • HTML • CSS
Laboratory	Electrochemical anodization • Spray pyrolysis • Hydrothermal synthesis • XRD • Raman spectroscopy • Photoluminescence spectroscopy • UV-Vis spectroscopy • Cleanroom protocols
Design	Adobe Photoshop • Adobe Illustrator • Canva • Social media management • Digital strategy
Languages	English (fluent) • Filipino (fluent) • Japanese (conversational) • German (beginner)
Interests	Photography • Editorial Writing • Fashion • Traveling • Fitness • Hiking